

# Qatar-1b

R-0608

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_230509 · created 2026-07-25T06:46:49+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460072.4257 +/- 0.0026 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.183 +/- 0.056                 |
| Transit depth (Rp/Rs)^2                           | 3.36 +/- 2.04 [%]               |
| Orbital Inclination (inc)                         | 79.37 +/- 2.86                  |
| Airmass coefficient 1 (a1)                        | 0.022 +/- 0.015                 |
| Airmass coefficient 2 (a2)                        | 2.1 +/- 1.53                    |
| Scatter in the residuals of the lightcurve fit is | 52.41 %                         |
| Best Comparison Star                              | #2 - [532, 442]                 |
| Optimal Aperture                                  | 2.65                            |
| Optimal Annulus                                   | 4.75                            |
| Transit Duration (day)                            | 0.0023 +/- 0.0045               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.183 $\pm$ 0.056 vs 0.1463 (deviation 0.66 $\sigma$ ) |
| Duration fit vs published            | 0.0023 d vs 0.0692 d (deviation 14.86 $\sigma$ )       |
| Mid-time O-C                         | -2049.9 min                                            |
| Depth SNR                            | 3.3                                                    |
| Residual scatter                     | 52.41 %                                                |

### Light curve

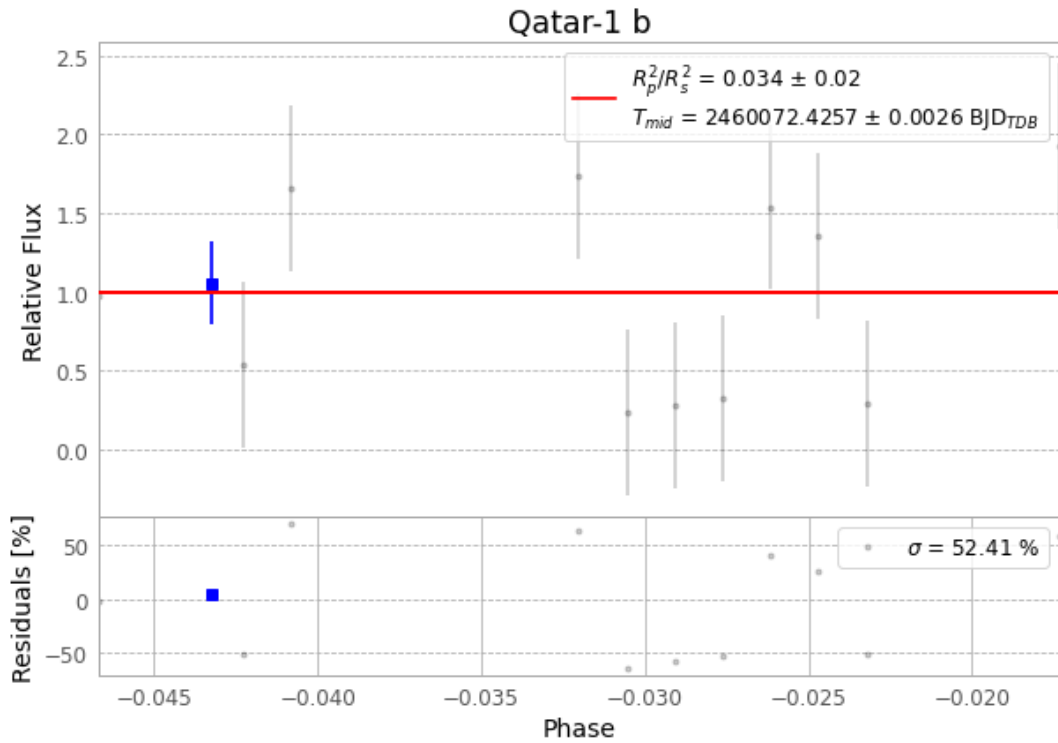


Figure 1 — FinalLightCurve\_Qatar-1 b\_2023-05-08.png (SHA-256 ec098d84b4776c67...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [461, 400], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 983d98ddf729d2e8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit b365494

## Data provenance

74 FITS frames (49 MB), Qatar-1230509063615.FITS → Qatar-1230509101515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230509.log (3e16353ec547...), FinalLightCurve\_Qatar-1 b\_2023-05-08.png (ec098d84b477...), FinalLightCurve\_Qatar-1 b\_2023-05-08.csv (87a26c9da54f...)

## Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 06:46Z.